VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

STUDENT RESULT AND GRADE MANAGEMENT SYSTEM

A9601 Object Oriented Programming through Java

Submitted by

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1 Problem Statement

Traditional notice boards are commonly used in schools, colleges, offices, and public places to communicate important information. However, updating these boards requires a person to write, print, and attach notices manually. This process consumes time and paper, and the information may not be updated immediately.

The proposed Real-Time Wireless Notice Board using Bluetooth provides a solution by allowing users to send messages wirelessly from a mobile phone to an electronic display. An Arduino Uno processes the received message and displays it on a 16x2 LCD through an HC-05 Bluetooth module. This reduces manual effort and makes notice updating easier and faster.

2 Existing System

In the existing system, notices are generally printed on paper or written manually and placed on a physical notice board. Whenever information changes, the old notice must be removed and a new notice must be displayed.

The main limitations of the existing system are:

- Frequent manual updating is required.
- Paper and printing costs may increase.
- Important information may not be displayed immediately.
- Old notices may remain on the board and cause confusion.
- Managing notices for multiple locations requires additional effort.

3 Proposed System

The proposed system uses Bluetooth communication to update an electronic notice board wirelessly. The user enters a text message in a compatible mobile application and sends it to the HC-05 Bluetooth module. The module transfers the received data to the Arduino Uno, which processes the message and sends the appropriate characters to the 16x2 LCD display.

The proposed system provides the following features:

- Wireless message transmission from a mobile device.
- Electronic display of text messages.
- Simple operation and easy message updating.
- Reduced dependence on printed notices.
- Low-cost implementation using commonly available components.

The basic Bluetooth implementation works within the effective range of the Bluetooth connection. Internet-based remote access is a possible future enhancement and is not provided by Bluetooth alone.

4 Scope of the Project

The scope of this project is to develop a small electronic notice board that can receive and display text messages through Bluetooth. It is suitable for demonstrations and small-scale applications where a user can update information from a nearby mobile device.

Possible applications include:

- Schools and colleges for classroom announcements.
- Offices for displaying short instructions.
- Laboratories for displaying safety messages.
- Shops for displaying short announcements.
- Reception areas for displaying temporary information.

The basic version is intended for short text messages on a 16x2 LCD. Longer messages may require scrolling or repeated display operations in the software.

5 Need for the System

A wireless notice board is useful when information must be updated regularly. Manual notice boards require physical access and repeated effort to replace notices. A Bluetooth-based electronic notice board allows a user to send a message from a mobile phone without directly touching the display unit.

The system also provides a practical way to understand embedded systems, serial communication, microcontroller programming, and LCD interfacing. It can be developed at a relatively low cost for educational demonstrations.

6 Working Principle

The system operates through the following steps:

1. The Arduino Uno is powered on and initializes the Bluetooth communication and LCD display.
2. The user enables Bluetooth on a compatible mobile phone and connects to the HC-05 module.
3. The user types a message in a compatible Bluetooth terminal application.
4. The mobile application sends the message wirelessly to the HC-05 Bluetooth module.
5. The HC-05 transfers the received serial data to the Arduino Uno.
6. The Arduino reads and processes the received characters.
7. The processed message is sent to the 16x2 LCD display.
8. The message appears on the LCD. If the message exceeds the display capacity, the program may scroll it or display it in sections.

7 Detailed Description of Components

7.1 Arduino Uno

The Arduino Uno is the main control unit of the system. It is based on the ATmega328P microcontroller and is responsible for receiving serial data, processing the message, and controlling the LCD display. The Arduino executes the program stored in its memory and coordinates communication between the system components.

7.2 HC-05 Bluetooth Module

The HC-05 is a Bluetooth serial communication module commonly used in educational electronics projects. It provides a wireless connection between the mobile phone and the Arduino. Text sent from the mobile application is received by the module and transferred to the microcontroller through serial communication.

The module supports short-range wireless communication. Its actual range depends on the environment, antenna, interference, and device configuration.

7.3 16x2 LCD Display

The 16x2 LCD is an electronic display capable of displaying 16 characters per line across two lines. It is used to show messages received from the mobile phone. The Arduino sends display commands and character data to the LCD.

Since the display has limited space, software may be required to scroll longer messages or show them in separate portions.

7.4 Potentiometer

A potentiometer is a variable resistor used to adjust the contrast of many character LCD modules. Proper contrast adjustment makes the displayed text easier to read. The potentiometer is generally connected to the LCD contrast control pin according to the circuit design.

7.5 Power Supply

The power supply provides the required operating voltage to the Arduino and connected modules. The components must be powered according to their electrical specifications. In particular, the HC-05 module's logic-level requirements should be checked, and the Arduino-to-module signal connection should use suitable voltage-level protection where necessary.

8 Software Requirements

The software requirements for the project include:

- **Arduino IDE:** Used to write, compile, and upload the program to the Arduino Uno.
- **Arduino Programming Language:** Used to control the microcontroller, receive serial data, and operate the LCD.
- **Bluetooth Terminal Application:** Used to send text messages from a compatible mobile phone.
- **LCD Library:** A suitable library, such as `LiquidCrystal`, may be used when the LCD is connected through a compatible parallel interface.

9 System Design and Data Flow

The system follows a simple input-processing-output model. The mobile phone acts as the input device, the HC-05 provides wireless communication, and the Arduino performs processing. The LCD serves as the output device.

1. **Input:** Text entered by the user on the mobile phone.
2. **Communication:** Transfer of text through Bluetooth.
3. **Processing:** Reading and preparing received characters using the Arduino program.
4. **Output:** Displaying the message on the LCD.

This modular design makes the system relatively easy to understand, test, and modify.

10 Advantages

The main advantages of the proposed system are:

- Reduces the effort required to update notices manually.
- Provides wireless message transmission.
- Uses affordable and commonly available components.
- Is suitable for educational demonstrations and small applications.
- Can be expanded with additional display and communication features.
- Helps reduce paper usage for frequently changing short messages.

11 Limitations

The basic system has certain limitations:

- Bluetooth communication works only within its effective range.
- The 16x2 LCD can display only a limited number of characters at a time.
- The mobile phone must be paired or connected appropriately.
- The system requires a suitable power supply.
- Bluetooth alone does not provide internet-based remote access.
- Access control should be considered if unauthorized users must be prevented from changing notices.

12 Testing and Verification

The system should be tested to confirm that messages are received and displayed correctly. The following test cases can be used during implementation.

These are proposed test cases. The actual results should be recorded after testing the assembled hardware and program.

Table 1: Test Cases for the Wireless Notice Board

Test	Input or Condition	Expected Result
1	Power on the system	Arduino initializes and the LCD becomes ready.
2	Connect the mobile phone using Bluetooth	The phone connects to the HC-05 module successfully.
3	Send a short text message	The message appears on the LCD.
4	Send a message longer than the display width	The program scrolls or displays the message in sections, depending on its implementation.
5	Disconnect Bluetooth	New messages are not received until communication is restored.

13 Future Enhancements

The system can be improved in several ways:

- **Wi-Fi Connectivity:** Add a Wi-Fi-enabled microcontroller or communication module to support network-based message updates.
- **Larger Display:** Use a larger LCD, LED matrix, or other display to show longer messages clearly.
- **Message Scrolling:** Implement automatic scrolling for messages that exceed the display capacity.
- **Access Control:** Add authentication or an approved sender mechanism to reduce unauthorized message changes.
- **Scheduled Messages:** Allow notices to be displayed at specified times.
- **Multiple Notice Boards:** Develop a network-based version that can send notices to multiple display units.

14 Conclusion

The Real-Time Wireless Notice Board using Bluetooth provides a simple and economical method of displaying text messages electronically. By combining an Arduino Uno, an HC-05 Bluetooth module, and a 16x2 LCD, the system allows a user to send messages from a compatible mobile phone without manually replacing a printed notice.

The proposed design can reduce the effort involved in routine notice updates and provides a useful practical application of microcontrollers and wireless communication. Although the basic implementation is limited by Bluetooth range and the capacity of the LCD, future improvements such as Wi-Fi connectivity, message scrolling, and access control can increase its functionality and usability. “‘ “‘latex

Figure 1 illustrates the architecture of the Wireless Notice Board system using an Arduino Uno, an HC-05 Bluetooth module, and a 16x2 LCD display. The user sends a text message through a compatible smartphone application. The HC-05 module receives the message and transfers it to the Arduino microcontroller through serial communication. The Arduino processes the received data and displays the message on the LCD screen. This system provides a

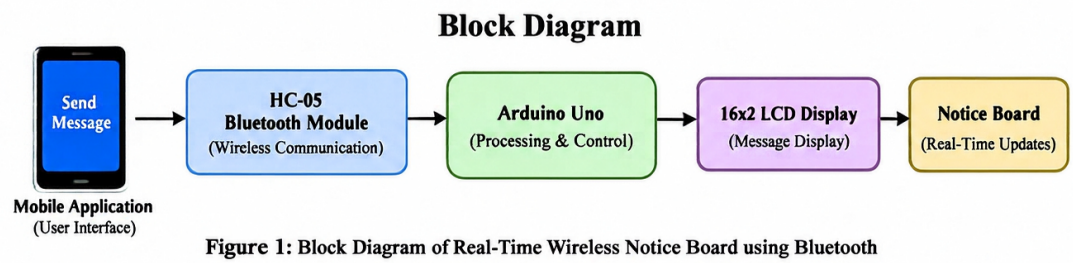


Figure 1: Block Diagram of Real-Time Wireless Notice Board using Bluetooth

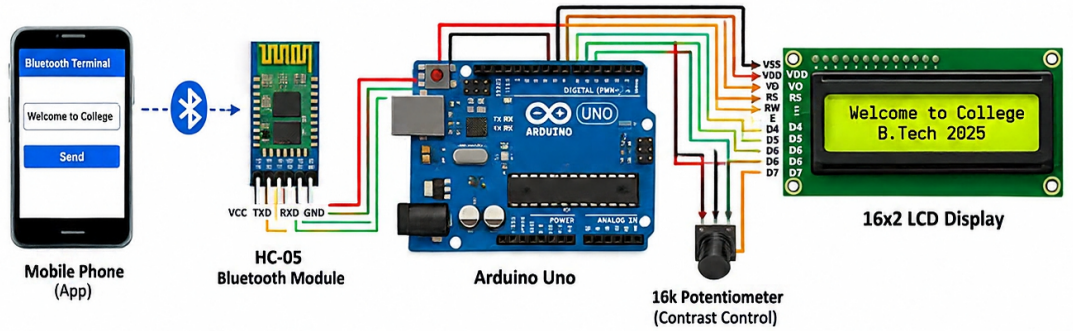


Figure 2: Circuit Diagram of Real-Time Wireless Notice Board using Bluetooth



Figure 3: Working Principle of Real-Time Wireless Notice Board

Figure 1: Wireless Notice Board Using Bluetooth

convenient method of updating notices wirelessly in educational institutions, offices, and other suitable locations.

14.1 Working Principle

The system operates according to the following steps:

1. The user pairs the smartphone with the HC-05 Bluetooth module.
2. A text message is entered into a compatible mobile application.
3. The message is transmitted wirelessly to the HC-05 module.
4. The Bluetooth module transfers the received data to the Arduino.
5. The Arduino processes the message and sends the characters to the 16x2 LCD display.
6. The message appears on the LCD. Longer messages can be displayed in sections or scrolled if the program supports this feature.

14.2 Software Implementation

The software components used in the project are described below:

- **Arduino IDE:** Used to write, compile, and upload the program to the Arduino Uno.
- **Arduino Programming Language:** Used to receive serial data, process messages, and control the LCD display.
- **Mobile Application:** MIT App Inventor or a compatible Serial Bluetooth Terminal application can be used to send text messages from a smartphone.

15 Results and Discussion

The proposed Wireless Notice Board is designed to display text messages received from a mobile device through Bluetooth communication. When a user sends a message, the HC-05 module receives the data and transfers it to the Arduino Uno. The Arduino processes the message and updates the 16x2 LCD display.

The system can reduce the time and effort required to update traditional paper-based notice boards. It also provides a simple and convenient way to display short announcements without physically replacing printed notices. The overall performance depends on the quality of the Bluetooth connection, the program implementation, and the power supply.

The communication range and response time should be measured during practical testing. The effective Bluetooth range depends on the environment, interference, and device configuration. The 16x2 LCD also limits the number of characters visible at one time; therefore, longer messages may require scrolling or multiple display sections.

A possible future enhancement is the integration of Wi-Fi connectivity. This would allow messages to be sent through a local network or an internet-connected application, depending on the design. Such an enhancement could support communication over greater distances than a direct Bluetooth connection.

Table 2: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applies microcontroller, electronic circuit, and communication principles.
PO2	Problem Analysis	Identifies the limitations of manual notice updating and the need for a wireless alternative.
PO3	Design/Development of Solutions	Develops a system for receiving and displaying text messages electronically.
PO5	Modern Tool Usage	Uses Arduino development tools, Bluetooth communication, and a mobile application.
PO6	Engineer and Society	Supports convenient information sharing in educational institutions and other suitable environments.

16 Mapping POs and SDGs

16.1 Program Outcomes (POs) Mapping

16.2 Sustainable Development Goals (SDGs) Mapping

Table 3: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	Goal	Relevance
SDG 9	Industry, Innovation and Infrastructure	Demonstrates the use of embedded systems and wireless communication to improve information display.
SDG 11	Sustainable Cities and Communities	Can support convenient information sharing in public and institutional spaces.
SDG 12	Responsible Consumption and Production	May reduce paper consumption when electronic messages replace frequently printed notices.

17 Conclusion

This project presents a Real-Time Wireless Notice Board using Bluetooth to simplify the process of updating and displaying short text messages. The system combines an Arduino Uno, an HC-05 Bluetooth module, and a 16x2 LCD display to receive messages from a compatible smartphone application and display them electronically.

The proposed design offers a simple and potentially cost-effective alternative to frequently updated paper notices. Its practical performance depends on Bluetooth connectivity, the LCD display capacity, and correct hardware implementation. Future improvements, including Wi-Fi connectivity, message scrolling, and access control, could increase the system's range, functionality, and security.

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